Mingxi Yan

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Education

Carnegie Mellon University *Expected August 2027*
Pittsburgh, PA
- Master of Science in Language Technologies (Artificial Intelligence).

Carnegie Mellon University *Graduated May 2026*
Pittsburgh, PA
- Bachelor of Science in Computer Science and Information Systems; Minor in Logic and Computation. **GPA**: 3.86 / 4.0

Skills

Programming: Python, SQL, C, JavaScript, TypeScript, Data Structures, Database Design, Distributed Systems
Machine Learning: PyTorch, vLLM, Hugging Face, TRL, PEFT/LoRA, SFT, DPO, Transformers, Multimodal ML
Agents: Agentic RAG, Agentic Search, tool use, LLM evaluation, Skills, MCP, Context Management
Data: PostgreSQL, SQLite, Docker, AWS, Feature Engineering, data pipelines, information retrieval, heuristic search

Industry Experience

Data Engineer Intern *June 2026 – Aug 2026*
TikTok, San Jose
- Designed a graph-based enterprise retrieval skill over Hive tables, datasets, dashboards, documents, and millions of metadata fields and values using query decomposition, contextual rewriting, graph expansion, and RRF.
- Built a benchmark using on-call Q&A pairs to evaluate agent results. Integrated as an internal published skill.
- Built a livestream revenue diagnosis copilot, using feature engineering, LightGBM percentile prediction, comparison cohorts, and business rules to detect revenue downside, identify abnormal metrics, retrieve relevant business documentation, and generate grounded recommendations. Validated with operations stakeholders and integrated into their workflow.

Full Stack Engineer & Tech Lead *Jun 2023 – Nov 2024*
Enterview (AI Startup), Pittsburgh, PA
- Led a four-engineer team building an AI interview-preparation platform while partnering on technical product decisions.
- Designed persistent user-context data models and backend APIs for personalized interview coaching, integrating LLM and Speech Recognition model to evaluate spoken responses and generate targeted feedback.

Research Experience

Research Assistant *May 2026 – Present*
Sapling Lab, Carnegie Mellon University, Pittsburgh, PA
- Lead NLP research for an NSF and OpenAI-funded interdisciplinary project studying how users form and update Theory-of-Mind attributions toward AI during long term human–AI interaction.
- Developing multi-turn user simulation to study user traits, assistant social behavior, memory, and scenarios while developing turn-level LLM judges and heuristic measures for evaluation.

Research Assistant *May 2024 – May 2026*
Chimps Lab, Carnegie Mellon University, Pittsburgh, PA
- Co-authored PersonaTeaming (UIST 2026) in collaboration with Apple Research, designing the persona-generation algorithm for adaptive LLM red-teaming. Evaluated 11 experimental conditions on 6 different LLM targets. Improved mean attack success rate by 46% and attack diversity by 7% over RainbowPlus (baseline). Paper.
- Built scalable image-generation bias-auditing infrastructure with persistent history, caching, batching, supporting 100 images per prompt. Designed interactive visualizations for researchers and participants to cooperate.

Research Intern *May 2023 – Aug 2023*
THUNLP Lab, Tsinghua University, Beijing — OpenBMB
- Converted REST APIs into LLM-callable tools with searchable metadata and instructions; built tests for tool invocation.
- Applied embedding-based clustering to categorize APIs and improve tool retrieval.

Selected Course Project

Solver-Guided Fine-Tuning for Poker Explanations *Spring 2026*
Carnegie Mellon University — Generative AI (10-423/623)
- Designed and implemented a solver-guided post-training pipeline for Llama 3 8B, constructing 3,000 SFT examples and 2,534 DPO pairs from PokerBench solver labels and verifier-ranked Llama 3.3 70B explanations.
- Implemented LoRA SFT/DPO and held-out evaluation, improving exact solver-action match from 18.0% to 56.2% and verifier score from 0.426 to 0.645 with SFT; identified an explanation-quality versus action-accuracy tradeoff under DPO.